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DIFFERENTIAL POWER AMPLIFIERS WITH COUPLED BIASING

Abstract

Apparatus and methods for biasing of differential power amplifiers are disclosed herein. In certain embodiments, a power amplifier includes a first gain transistor for amplifying a first signal component of a radio frequency (RF) signal and a second gain transistor for amplifying a second signal component of the RF signal. The power amplifier further includes a shared bias circuit for generating a bias signal for biasing an input of the first gain transistor and an input of the second gain transistor. The RF signal is coupled into the shared bias circuit to enhance biasing performance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Patent Application No. 63/644,670, filed May 9, 2024 and titled “DIFFERENTIAL POWER AMPLIFIERS WITH COUPLED BIASING,” and of U.S. Provisional Patent Application No. 63/560,972, filed Mar. 4, 2024 and titled “BIASING OF DIFFERENTIAL POWER AMPLIFIERS,” each of which is herein incorporated by reference in its entirety.

BACKGROUND

Field

[0002] Embodiments of the invention relate to electronic systems, and in particular, to radio frequency electronics.

Description of Related Technology

[0003] Radio frequency (RF) communication systems can be used for transmitting and/or receiving signals of a wide range of frequencies. For example, an RF communication system can be used to wirelessly communicate RF signals in a frequency range of about 30 kHz to 300 GHz, such as in the range of about 410 MHz to about 7.125 GHz for Fifth Generation (5G) cellular communications in Frequency Range 1 (FR1) or in the range of about 24.250 GHz to about 71.000 GHz for Frequency Range 2 (FR2) of the 5G communication standard.

[0004] Examples of RF communication systems include, but are not limited to, mobile phones, tablets, base stations, network access points, customer-premises equipment (CPE), laptops, and/or wearable electronics.

SUMMARY

[0005] In certain embodiments, the present disclosure relates to a mobile device. The mobile device includes a transceiver configured to generate a radio frequency input signal, and a front-end system including a power amplifier that includes a first amplification device configured to amplify a first signal component of a radio frequency input signal, a second amplification device configured to amplify a second signal component of the radio frequency input signal, a shared bias circuit having a first output configured to generate a first bias signal and a second output configured to generate a second bias signal, a first coupling impedance connected between the first output of the shared bias circuit and an input of the first amplification device, and a second coupling impedance connected between the second output of the shared bias circuit and an input of the second amplification device. The first coupling impedance is configured to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output, and the second coupling impedance is configured to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.

[0006] In various embodiments, the power amplifier further includes an input terminal configured to receive the radio frequency input signal. According to a number of embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. In accordance with several embodiments, the power amplifier further includes a first capacitor connected between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second capacitor connected between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0007] In some embodiments, the shared bias circuit generates the first bias signal and the second

bias signal based on a common reference signal.

[0008] In various embodiments, the first coupling impedance includes a first resistor and a first capacitor connected in parallel between the first output of the shared bias circuit and the input of the first amplification device, and the second coupling impedance includes a second resistor and a second capacitor connected in parallel between the second output of the shared bias circuit and the input of the second amplification device.

[0009] In several embodiments, the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared bias circuit. According to a number of embodiments, a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.

[0010] In some embodiments, the first amplification device includes a first bipolar transistor having a base corresponding to the input of the first amplification device, and the second amplification device includes a second bipolar transistor having a base corresponding to the input of the second amplification device.

[0011] In various embodiments, the front end system includes a Doherty power amplifier system including the power amplifier. According to several embodiments, the power amplifier is one of a peaking amplifier or a carrier amplifier of the Doherty power amplifier system.

[0012] In certain embodiments, the present disclosure relates to a power amplifier. The power amplifier includes a first amplification device configured to amplify a first signal component of a radio frequency input signal, a second amplification device configured to amplify a second signal component of the radio frequency input signal, a shared bias circuit having a first output configured to generate a first bias signal and a second output configured to generate a second bias signal, a first coupling impedance connected between the first output of the shared bias circuit and an input of the first amplification device, and a second coupling impedance connected between the second output of the shared bias circuit and an input of the second amplification device. The first coupling impedance is configured to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output, and the second coupling impedance is configured to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.

[0013] In various embodiments, the power amplifier further includes an input terminal configured to receive the radio frequency input signal. According to a number of embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. In accordance with several embodiments, the power amplifier further includes a first capacitor connected between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second capacitor connected between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0014] In some embodiments, the shared bias circuit generates the first bias signal and the second bias signal based on a common reference signal.

[0015] In various embodiments, the first coupling impedance includes a first resistor and a first capacitor connected in parallel between the first output of the shared bias circuit and the input of the first amplification device, and the second coupling impedance includes a second resistor and a second capacitor connected in parallel between the second output of the shared bias circuit and the input of the second amplification device.

[0016] In several embodiments, the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared

bias circuit. According to a number of embodiments, a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.

[0017] In various embodiments, the first amplification device includes a first bipolar transistor having a base corresponding to the input of the first amplification device, and the second amplification device includes a second bipolar transistor having a base corresponding to the input of the second amplification device.

[0018] In certain embodiments, the present disclosure relates to a method of radio frequency signal amplification in a mobile device. The method includes amplifying a first signal component of a radio frequency input signal using a first amplification device of a power amplifier, amplifying a second signal component of the radio frequency input signal using a second amplification device of the power amplifier, generating a first bias signal at a first output of a shared bias circuit and a second bias signal at a second output of the shared bias circuit, using a first coupling impedance to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output, and using a second coupling impedance to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.

[0019] In various embodiments, the power amplifier further includes an input terminal configured to receive the radio frequency input signal. According to a number of embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. In accordance with some embodiments, the power amplifier further includes a first capacitor connected between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second capacitor connected between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0020] In several embodiments, the shared bias circuit generates the first bias signal and the second bias signal based on a common reference signal.

[0021] In some embodiments, the first coupling impedance includes a first resistor and a first capacitor connected in parallel between the first output of the shared bias circuit and the input of the first amplification device, and the second coupling impedance includes a second resistor and a second capacitor connected in parallel between the second output of the shared bias circuit and the input of the second amplification device.

[0022] In several embodiments, the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared bias circuit. According to a number of embodiments, a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.

[0023] In various embodiments, the first amplification device includes a first bipolar transistor having a base corresponding to the input of the first amplification device, and the second amplification device includes a second bipolar transistor having a base corresponding to the input of the second amplification device.

[0024] In certain embodiments, the present disclosure relates to a power amplifier. The power amplifier includes an input terminal configured to receive a radio frequency input signal, a first amplification device configured to amplify a first signal component of the radio frequency input signal, a second amplification device configured to amplify a second signal component of the radio frequency input signal, and a shared bias circuit having an output configured to generate a bias signal for biasing an input of the first amplification device and an input of the second amplification device. The input terminal is coupled to the output of the shared bias circuit to compensate the bias

signal for changes in power of the radio frequency input signal.

[0025] In some embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit.

[0026] In several embodiments, the power amplifier further includes a first biasing impedance connected between the output of the shared bias circuit and the input of the first amplification device and a second biasing impedance connected between the output of the shared bias circuit and the input of the second amplification device.

[0027] In some embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. According to a number of embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit. In accordance with several embodiments, the power amplifier further includes a biasing inductor connected between a center tap of the secondary winding and the output of the shared bias circuit. According to various embodiments, the power amplifier further includes a coupling inductor connected between a center tap of the secondary winding and the output of the shared bias circuit, the coupling inductor magnetically coupled to the input balun. In accordance with a number of the embodiments, the power amplifier further includes a first resistor and a first capacitor connected in parallel between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second resistor and a second capacitor in parallel between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0028] In some embodiments, the shared bias circuit includes an emitter follower bipolar transistor having an emitter connected to the output of the shared bias circuit.

[0029] In various embodiments, the first amplification device includes a first bipolar transistor having a base corresponding to the input of the first amplification device, and the second amplification device includes a second bipolar transistor having a base corresponding to the input of the second amplification device.

[0030] In certain embodiments, the present disclosure relates to a mobile device. The mobile device includes a transceiver configured to generate a radio frequency input signal, and a front-end system including a power amplifier that includes an input terminal configured to receive the radio frequency input signal, a first amplification device configured to amplify a first signal component of the radio frequency input signal, a second amplification device configured to amplify a second signal component of the radio frequency input signal, and a shared bias circuit having an output configured to generate a bias signal for biasing an input of the first amplification device and an input of the second amplification device. The input terminal is coupled to the output of the shared bias circuit to compensate the bias signal for changes in power of the radio frequency input signal.

[0031] In various embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit.

[0032] In some embodiments, the power amplifier further includes a first biasing impedance connected between the output of the shared bias circuit and the input of the first amplification device and a second biasing impedance connected between the output of the shared bias circuit and the input of the second amplification device.

[0033] In several embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. According to a number of embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit. In accordance with some embodiments, the power amplifier further includes a biasing inductor connected between a center tap of the secondary winding and the output of the shared bias circuit. According to various embodiments, the power amplifier further includes a coupling inductor connected between a center

tap of the secondary winding and the output of the shared bias circuit, the coupling inductor magnetically coupled to the input balun. In accordance with a number of embodiments, the power amplifier further includes a first resistor and a first capacitor connected in parallel between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second resistor and a second capacitor connected in parallel between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0034] In some embodiments, the shared bias circuit includes an emitter follower bipolar transistor having an emitter connected to the output of the shared bias circuit.

[0035] In several embodiments, the front end system includes a Doherty power amplifier system including the power amplifier. According to a number of embodiments, the power amplifier is one of a peaking amplifier or a carrier amplifier of the Doherty power amplifier system.

[0036] In certain embodiments, the present disclosure relates to a method of radio frequency signal amplification in a mobile device. The method includes receiving a radio frequency input signal at an input terminal of a power amplifier, amplifying a first signal component of the radio frequency input signal using a first amplification device of the power amplifier, amplifying a second signal component of the radio frequency input signal using a second amplification device of the power amplifier, generating a bias signal for biasing an input of the first amplification device and an input of the second amplification device at an output of a shared bias circuit of the power amplifier, and compensating the bias signal for changes in power of the radio frequency input signal by providing coupling from the input terminal to the output of the shared bias circuit.

[0037] In some embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit.

[0038] In several embodiments, the power amplifier further includes a first biasing impedance connected between the output of the shared bias circuit and the input of the first amplification device and a second biasing impedance connected between the output of the shared bias circuit and the input of the second amplification device.

[0039] In some embodiments, the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device. According to a number of embodiments, the power amplifier further includes a coupling capacitor connected between the input terminal and the output of the shared bias circuit. In accordance with several embodiments, the power amplifier further includes a biasing inductor connected between a center tap of the secondary winding and the output of the shared bias circuit. According to various embodiments, the power amplifier further includes a coupling inductor connected between a center tap of the secondary winding and the output of the shared bias circuit, the coupling inductor magnetically coupled to the input balun. In accordance with a number of embodiments, the power amplifier further includes a first resistor and a first capacitor connected in parallel between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second resistor and a second capacitor connected in parallel between a second end of the secondary winding of the input balun and the input of the second amplification device.

[0040] In various embodiments, the shared bias circuit includes an emitter follower bipolar transistor having an emitter connected to the output of the shared bias circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] FIG. 1 is a schematic diagram of one example of a communication network.

[0042] FIG. 2A is a schematic diagram of one example of a communication link using carrier aggregation.

[0043] FIG. 2B illustrates various examples of uplink carrier aggregation for the communication link of FIG. 2A.

[0044] FIG. 2C illustrates various examples of downlink carrier aggregation for the communication link of FIG. 2A.

[0045] FIG. 3A is a schematic diagram of one example of a downlink channel using multi-input and multi-output (MIMO) communications.

[0046] FIG. 3B is schematic diagram of one example of an uplink channel using MIMO communications.

[0047] FIG. 3C is schematic diagram of another example of an uplink channel using MIMO communications.

[0048] FIG. 4 is a schematic diagram of an example dual connectivity network topology.

[0049] FIG. 5A is a schematic diagram of a power amplifier according to one embodiment.

[0050] FIG. 5B is a schematic diagram of a power amplifier according to another embodiment.

[0051] FIG. 5C is a schematic diagram of a power amplifier according to another embodiment.

[0052] FIG. 6 is a schematic diagram of a power amplifier according to another embodiment.

[0053] FIG. 7 is a schematic diagram of a Doherty power amplifier system according to one embodiment.

[0054] FIG. 8A is a schematic diagram of a power amplifier according to another embodiment.

[0055] FIG. 8B is a schematic diagram of a power amplifier according to another embodiment.

[0056] FIG. 8C is a schematic diagram of a power amplifier according to another embodiment.

[0057] FIG. 9A is a schematic diagram of a power amplifier according to another embodiment.

[0058] FIG. 9B is a schematic diagram of a power amplifier according to another embodiment.

[0059] FIG. 10A depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier of FIG. 8A.

[0060] FIG. 10B depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier of FIG. 8B.

[0061] FIG. 10C depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier of FIG. 8C.

[0062] FIG. 11A depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier of FIG. 9A.

[0063] FIG. 11B depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier of FIG. 9B.

[0064] FIG. 12 is a schematic diagram of one embodiment of a mobile device.

[0065] FIG. 13 is a schematic diagram of a power amplifier system according to one embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0066] The following detailed description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

[0067] The International Telecommunication Union (ITU) is a specialized agency of the United Nations (UN) responsible for global issues concerning information and communication technologies, including the shared global use of radio spectrum.

[0068] The 3rd Generation Partnership Project (3GPP) is a collaboration between groups of telecommunications standard bodies across the world, such as the Association of Radio Industries and Businesses (ARIB), the Telecommunications Technology Committee (TTC), the China Communications Standards Association (CCSA), the Alliance for Telecommunications Industry

Solutions (ATIS), the Telecommunications Technology Association (TTA), the European Telecommunications Standards Institute (ETSI), and the Telecommunications Standards Development Society, India (TSDSI).

[0069] Working within the scope of the ITU, 3GPP develops and maintains technical specifications for a variety of mobile communication technologies, including, for example, second generation (2G) technology (for instance, Global System for Mobile Communications (GSM) and Enhanced Data Rates for GSM Evolution (EDGE)), third generation (3G) technology (for instance, Universal Mobile Telecommunications System (UMTS) and High Speed Packet Access (HSPA)), and fourth generation (4G) technology (for instance, Long Term Evolution (LTE) and LTE-Advanced).

[0070] The technical specifications controlled by 3GPP can be expanded and revised by specification releases, which can span multiple years and specify a breadth of new features and evolutions.

[0071] In one example, 3GPP introduced carrier aggregation (CA) for LTE in Release 10. Although initially introduced with two downlink carriers, 3GPP expanded carrier aggregation in Release 14 to include up to five downlink carriers and up to three uplink carriers. Other examples of new features and evolutions provided by 3GPP releases include, but are not limited to, License Assisted Access (LAA), enhanced LAA (eLAA), Narrowband Internet of things (NB-IOT), Vehicle-to-Everything (V2X), and High Power User Equipment (HPUE).

[0072] 3GPP introduced Phase 1 of fifth generation (5G) technology in Release 15 and introduced Phase 2 of 5G technology in Release 16. Subsequent 3GPP releases will further evolve and expand 5G technology. 5G technology is also referred to herein as 5G New Radio (NR).

[0073] 5G NR supports or plans to support a variety of features, such as communications over millimeter wave spectrum, beamforming capability, high spectral efficiency waveforms, low latency communications, multiple radio numerology, and/or non-orthogonal multiple access (NOMA). Although such RF functionalities offer flexibility to networks and enhance user data rates, supporting such features can pose a number of technical challenges.

[0074] The teachings herein are applicable to a wide variety of communication systems, including, but not limited to, communication systems using advanced cellular technologies, such as LTE-Advanced, LTE-Advanced Pro, and/or 5G NR.

[0075] FIG. 1 is a schematic diagram of one example of a communication network **10**. The communication network **10** includes a macro cell base station **1**, a small cell base station **3**, and various examples of user equipment (UE), including a first mobile device **2a**, a wireless-connected car **2b**, a laptop **2c**, a stationary wireless device **2d**, a wireless-connected train **2e**, a second mobile device **2f**, and a third mobile device **2g**.

[0076] Although specific examples of base stations and user equipment are illustrated in FIG. 1, a communication network can include base stations and user equipment of a wide variety of types and/or numbers.

[0077] For instance, in the example shown, the communication network **10** includes the macro cell base station **1** and the small cell base station **3**. The small cell base station **3** can operate with relatively lower power, shorter range, and/or with fewer concurrent users relative to the macro cell base station **1**. The small cell base station **3** can also be referred to as a femtocell, a picocell, or a microcell. Although the communication network **10** is illustrated as including two base stations, the communication network **10** can be implemented to include more or fewer base stations and/or base stations of other types.

[0078] Although various examples of user equipment are shown, the teachings herein are applicable to a wide variety of user equipment, including, but not limited to, mobile phones, tablets, laptops, IoT devices, wearable electronics, customer premises equipment (CPE), wireless-connected vehicles, wireless relays, and/or a wide variety of other communication devices. Furthermore, user equipment includes not only currently available communication devices that operate in a cellular network, but also subsequently developed communication devices that will be

readily implementable with the inventive systems, processes, methods, and devices as described and claimed herein.

[0079] The illustrated communication network **10** of FIG. **1** supports communications using a variety of cellular technologies, including, for example, 4G LTE and 5G NR. In certain implementations, the communication network **10** is further adapted to provide a wireless local area network (WLAN), such as Wi-Fi®. Although various examples of communication technologies have been provided, the communication network **10** can be adapted to support a wide variety of communication technologies.

[0080] Various communication links of the communication network **10** have been depicted in FIG. **1**. The communication links can be duplexed in a wide variety of ways, including, for example, using frequency-division duplexing (FDD) and/or time-division duplexing (TDD). FDD is a type of radio frequency communications that uses different frequencies for transmitting and receiving signals. FDD can provide a number of advantages, such as high data rates and low latency. In contrast, TDD is a type of radio frequency communications that uses about the same frequency for transmitting and receiving signals, and in which transmit and receive communications are switched in time. TDD can provide a number of advantages, such as efficient use of spectrum and variable allocation of throughput between transmit and receive directions.

[0081] In certain implementations, user equipment can communicate with a base station using one or more of 4G LTE, 5G NR, and Wi-Fi® technologies. In certain implementations, enhanced license assisted access (eLAA) is used to aggregate one or more licensed frequency carriers (for instance, licensed 4G LTE and/or 5G NR frequencies), with one or more unlicensed carriers (for instance, unlicensed Wi-Fi® frequencies).

[0082] As shown in FIG. **1**, the communication links include not only communication links between UE and base stations, but also UE to UE communications and base station to base station communications. For example, the communication network **10** can be implemented to support self-fronthaul and/or self-backhaul (for instance, as between mobile device **2g** and mobile device **2f**).

[0083] The communication links can operate over a wide variety of frequencies. In certain implementations, communications are supported using 5G NR technology over one or more frequency bands that are less than 6 Gigahertz (GHz) and/or over one or more frequency bands that are greater than 6 GHz. For example, the communication links can serve Frequency Range 1 (FR1), Frequency Range 2 (FR2), or a combination thereof. In one embodiment, one or more of the mobile devices support a HPUE power class specification.

[0084] In certain implementations, a base station and/or user equipment communicates using beamforming. For example, beamforming can be used to focus signal strength to overcome path losses, such as high loss associated with communicating over high signal frequencies. In certain embodiments, user equipment, such as one or more mobile phones, communicate using beamforming on millimeter wave frequency bands in the range of 30 GHz to 300 GHz and/or upper centimeter wave frequencies in the range of 6 GHz to 30 GHz, or more particularly, 24 GHz to 30 GHz. Cellular user equipment can communicate using beamforming and/or other techniques over a wide range of frequencies, including, for example, FR2-1 (24 GHz to 52 GHz), FR2-2 (52 GHz to 71 GHz), and/or FR1 (400 MHz to 7125 MHz).

[0085] Different users of the communication network **10** can share available network resources, such as available frequency spectrum, in a wide variety of ways.

[0086] In one example, frequency division multiple access (FDMA) is used to divide a frequency band into multiple frequency carriers. Additionally, one or more carriers are allocated to a particular user. Examples of FDMA include, but are not limited to, single carrier FDMA (SC-FDMA) and orthogonal FDMA (OFDMA). OFDMA is a multicarrier technology that subdivides the available bandwidth into multiple mutually orthogonal narrowband subcarriers, which can be separately assigned to different users.

[0087] Other examples of shared access include, but are not limited to, time division multiple

access (TDMA) in which a user is allocated particular time slots for using a frequency resource, code division multiple access (CDMA) in which a frequency resource is shared amongst different users by assigning each user a unique code, space-divisional multiple access (SDMA) in which beamforming is used to provide shared access by spatial division, and non-orthogonal multiple access (NOMA) in which the power domain is used for multiple access. For example, NOMA can be used to serve multiple users at the same frequency, time, and/or code, but with different power levels.

[0088] Enhanced mobile broadband (eMBB) refers to technology for growing system capacity of LTE networks. For example, eMBB can refer to communications with a peak data rate of at least 10 Gbps and a minimum of 100 Mbps for each user. Ultra-reliable low latency communications (uRLLC) refers to technology for communication with very low latency, for instance, less than 2 milliseconds. uRLLC can be used for mission-critical communications such as for autonomous driving and/or remote surgery applications. Massive machine-type communications (mMTC) refers to low cost and low data rate communications associated with wireless connections to everyday objects, such as those associated with Internet of Things (IoT) applications.

[0089] The communication network **10** of FIG. **1** can be used to support a wide variety of advanced communication features, including, but not limited to, eMBB, uRLLC, and/or mMTC.

[0090] In certain implementations, the communication network **10** supports supplementary uplink (SUL) and/or supplementary downlink (SDL). For example, when channel conditions are good, the communication network **10** can direct a particular UE to transmit using an original uplink frequency, while when channel condition is poor (for instance, below a certain criteria) the communication network **10** can direct the UE to transmit using a supplementary uplink frequency that is lower than the original uplink frequency. Since cell coverage increases with lower frequency, communication range and/or signal-to-noise ratio (SNR) can be increased using SUL. Likewise, SDL can be used to transmit using an original downlink frequency when channel conditions are good, and to transmit using a supplementary downlink frequency when channel conditions are poor.

[0091] FIG. **2A** is a schematic diagram of one example of a communication link using carrier aggregation. Carrier aggregation can be used to widen bandwidth of the communication link by supporting communications over multiple frequency carriers, thereby increasing user data rates and enhancing network capacity by utilizing fragmented spectrum allocations.

[0092] In the illustrated example, the communication link is provided between a base station **21** and a mobile device **22**. As shown in FIG. **2A**, the communications link includes a downlink channel used for RF communications from the base station **21** to the mobile device **22**, and an uplink channel used for RF communications from the mobile device **22** to the base station **21**.

[0093] Although FIG. **2A** illustrates carrier aggregation in the context of FDD communications, carrier aggregation can also be used for TDD communications.

[0094] In certain implementations, a communication link can provide asymmetrical data rates for a downlink channel and an uplink channel. For example, a communication link can be used to support a relatively high downlink data rate to enable high speed streaming of multimedia content to a mobile device, while providing a relatively slower data rate for uploading data from the mobile device to the cloud.

[0095] In the illustrated example, the base station **21** and the mobile device **22** communicate via carrier aggregation, which can be used to selectively increase bandwidth of the communication link. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

[0096] In the example shown in FIG. **2A**, the uplink channel includes three aggregated component carriers f.sub.UL1, f.sub.UL2, and f.sub.UL3. Additionally, the downlink channel includes five aggregated component carriers f.sub.DL1, f.sub.DL2, f.sub.DL3, f.sub.DL4, and f.sub.DL5.

Although one example of component carrier aggregation is shown, more or fewer carriers can be aggregated for uplink and/or downlink. Moreover, a number of aggregated carriers can be varied over time to achieve desired uplink and downlink data rates.

[0097] For example, a number of aggregated carriers for uplink and/or downlink communications with respect to a particular mobile device can change over time. For example, the number of aggregated carriers can change as the device moves through the communication network and/or as network usage changes over time.

[0098] FIG. 2B illustrates various examples of uplink carrier aggregation for the communication link of FIG. 2A. FIG. 2B includes a first carrier aggregation scenario **31**, a second carrier aggregation scenario **32**, and a third carrier aggregation scenario **33**, which schematically depict three types of carrier aggregation.

[0099] The carrier aggregation scenarios **31-33** illustrate different spectrum allocations for a first component carrier $f_{\text{sub}}\text{UL1}$, a second component carrier $f_{\text{sub}}\text{UL2}$, and a third component carrier $f_{\text{sub}}\text{UL3}$. Although FIG. 2B is illustrated in the context of aggregating three component carriers, carrier aggregation can be used to aggregate more or fewer carriers. Moreover, although illustrated in the context of uplink, the aggregation scenarios are also applicable to downlink.

[0100] The first carrier aggregation scenario **31** illustrates intra-band contiguous carrier aggregation, in which component carriers that are adjacent in frequency and in a common frequency band are aggregated. For example, the first carrier aggregation scenario **31** depicts aggregation of component carriers $f_{\text{sub}}\text{UL1}$, $f_{\text{sub}}\text{UL2}$, and $f_{\text{sub}}\text{UL3}$ that are contiguous and located within a first frequency band **BAND1**.

[0101] With continuing reference to FIG. 2B, the second carrier aggregation scenario **32** illustrates intra-band non-continuous carrier aggregation, in which two or more components carriers that are non-adjacent in frequency and within a common frequency band are aggregated. For example, the second carrier aggregation scenario **32** depicts aggregation of component carriers $f_{\text{sub}}\text{UL1}$, $f_{\text{sub}}\text{UL2}$, and $f_{\text{sub}}\text{UL3}$ that are non-contiguous, but located within a first frequency band **BAND1**.

[0102] The third carrier aggregation scenario **33** illustrates inter-band non-contiguous carrier aggregation, in which component carriers that are non-adjacent in frequency and in multiple frequency bands are aggregated. For example, the third carrier aggregation scenario **33** depicts aggregation of component carriers $f_{\text{sub}}\text{UL1}$ and $f_{\text{sub}}\text{UL2}$ of a first frequency band **BAND1** with component carrier $f_{\text{sub}}\text{UL3}$ of a second frequency band **BAND2**.

[0103] FIG. 2C illustrates various examples of downlink carrier aggregation for the communication link of FIG. 2A. The examples depict various carrier aggregation scenarios **34-38** for different spectrum allocations of a first component carrier $f_{\text{sub}}\text{DL1}$, a second component carrier $f_{\text{sub}}\text{DL2}$, a third component carrier $f_{\text{sub}}\text{DL3}$, a fourth component carrier $f_{\text{sub}}\text{DL4}$, and a fifth component carrier $f_{\text{sub}}\text{DL5}$. Although FIG. 2C is illustrated in the context of aggregating five component carriers, carrier aggregation can be used to aggregate more or fewer carriers. Moreover, although illustrated in the context of downlink, the aggregation scenarios are also applicable to uplink.

[0104] The first carrier aggregation scenario **34** depicts aggregation of component carriers that are contiguous and located within the same frequency band. Additionally, the second carrier aggregation scenario **35** and the third carrier aggregation scenario **36** illustrates two examples of aggregation that are non-contiguous, but located within the same frequency band. Furthermore, the fourth carrier aggregation scenario **37** and the fifth carrier aggregation scenario **38** illustrates two examples of aggregation in which component carriers that are non-adjacent in frequency and in multiple frequency bands are aggregated. As a number of aggregated component carriers increases, a complexity of possible carrier aggregation scenarios also increases.

[0105] With reference to FIGS. 2A-2C, the individual component carriers used in carrier aggregation can be of a variety of frequencies, including, for example, frequency carriers in the same band or in multiple bands. Additionally, carrier aggregation is applicable to implementations

in which the individual component carriers are of about the same bandwidth as well as to implementations in which the individual component carriers have different bandwidths.

[0106] Certain communication networks allocate a particular user device with a primary component carrier (PCC) or anchor carrier for uplink and a PCC for downlink. Additionally, when the mobile device communicates using a single frequency carrier for uplink or downlink, the user device communicates using the PCC. To enhance bandwidth for uplink communications, the uplink PCC can be aggregated with one or more uplink secondary component carriers (SCCs).

Additionally, to enhance bandwidth for downlink communications, the downlink PCC can be aggregated with one or more downlink SCCs.

[0107] In certain implementations, a communication network provides a network cell for each component carrier. Additionally, a primary cell can operate using a PCC, while a secondary cell can operate using a SCC. The primary and secondary cells may have different coverage areas, for instance, due to differences in frequencies of carriers and/or network environment.

[0108] License assisted access (LAA) refers to downlink carrier aggregation in which a licensed frequency carrier associated with a mobile operator is aggregated with a frequency carrier in unlicensed spectrum, such as Wi-Fi®. LAA employs a downlink PCC in the licensed spectrum that carries control and signaling information associated with the communication link, while unlicensed spectrum is aggregated for wider downlink bandwidth when available. LAA can operate with dynamic adjustment of secondary carriers to avoid Wi-Fi® users and/or to coexist with Wi-Fi® users. Enhanced license assisted access (eLAA) refers to an evolution of LAA that aggregates licensed and unlicensed spectrum for both downlink and uplink. Furthermore, NR-U can operate on top of LAA/eLAA over a 5 GHz band (5150 to 5925 MHz) and/or a 6 GHz band (5925 MHz to 7125 MHz).

[0109] FIG. 3A is a schematic diagram of one example of a downlink channel using multi-input and multi-output (MIMO) communications. FIG. 3B is schematic diagram of one example of an uplink channel using MIMO communications.

[0110] MIMO communications use multiple antennas for simultaneously communicating multiple data streams over common frequency spectrum. In certain implementations, the data streams operate with different reference signals to enhance data reception at the receiver. MIMO communications benefit from higher SNR, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment.

[0111] MIMO order refers to a number of separate data streams sent or received. For instance, MIMO order for downlink communications can be described by a number of transmit antennas of a base station and a number of receive antennas for UE, such as a mobile device. For example, two-by-two (2×2) DL MIMO refers to MIMO downlink communications using two base station antennas and two UE antennas. Additionally, four-by-four (4×4) DL MIMO refers to MIMO downlink communications using four base station antennas and four UE antennas.

[0112] In the example shown in FIG. 3A, downlink MIMO communications are provided by transmitting using M antennas 43a, 43b, 43c, . . . 43m of the base station 41 and receiving using N antennas 44a, 44b, 44c, . . . 44n of the mobile device 42. Accordingly, FIG. 3A illustrates an example of m×n DL MIMO.

[0113] Likewise, MIMO order for uplink communications can be described by a number of transmit antennas of UE, such as a mobile device, and a number of receive antennas of a base station. For example, 2×2 UL MIMO refers to MIMO uplink communications using two UE antennas and two base station antennas. Additionally, 4×4 UL MIMO refers to MIMO uplink communications using four UE antennas and four base station antennas.

[0114] In the example shown in FIG. 3B, uplink MIMO communications are provided by transmitting using N antennas 44a, 44b, 44c, . . . 44n of the mobile device 42 and receiving using M antennas 43a, 43b, 43c, . . . 43m of the base station 41. Accordingly, FIG. 3B illustrates an example of n×m UL MIMO.

[0115] By increasing the level or order of MIMO, bandwidth of an uplink channel and/or a downlink channel can be increased.

[0116] MIMO communications are applicable to communication links of a variety of types, such as FDD communication links and TDD communication links.

[0117] FIG. 3C is schematic diagram of another example of an uplink channel using MIMO communications. In the example shown in FIG. 3C, uplink MIMO communications are provided by transmitting using N antennas **44a**, **44b**, **44c**, . . . **44n** of the mobile device **42**. Additionally a first portion of the uplink transmissions are received using M antennas **43a1**, **43b1**, **43c1**, . . . **43m1** of a first base station **41a**, while a second portion of the uplink transmissions are received using M antennas **43a2**, **43b2**, **43c2**, . . . **43m2** of a second base station **41b**. Additionally, the first base station **41a** and the second base station **41b** communicate with one another over wired, optical, and/or wireless links.

[0118] The MIMO scenario of FIG. 3C illustrates an example in which multiple base stations cooperate to facilitate MIMO communications.

[0119] With the introduction of the 5G NR air interface standards, 3GPP has allowed for the simultaneous operation of 5G and 4G standards in order to facilitate the transition. This mode can be referred to as Non-Stand-Alone (NSA) operation or E-UTRAN New Radio-Dual Connectivity (EN-DC) and involves both 4G and 5G carriers being simultaneously transmitted from user equipment (UE).

[0120] In certain EN-DC applications, dual connectivity NSA involves overlaying 5G systems onto an existing 4G core network. For dual connectivity in such applications, the control and synchronization between the base station and the UE can be performed by the 4G network while the 5G network is a complementary radio access network tethered to the 4G anchor. The 4G anchor can connect to the existing 4G network with the overlay of 5G data/control.

[0121] FIG. 4 is a schematic diagram of an example dual connectivity network topology. This architecture can leverage LTE legacy coverage to ensure continuity of service delivery and the progressive rollout of 5G cells. A UE **2** can simultaneously transmit dual uplink LTE and NR carrier. The UE **2** can transmit an uplink LTE carrier Tx1 to the eNB **11** while transmitting an uplink NR carrier Tx2 to the gNB **12** to implement dual connectivity. Any suitable combination of uplink carriers Tx1, Tx2 and/or downlink carriers Rx1, Rx2 can be concurrently transmitted via wireless links in the example network topology of FIG. 1. The eNB **11** can provide a connection with a core network, such as an Evolved Packet Core (EPC) **14**. The gNB **12** can communicate with the core network via the eNB **11**. Control plane data can be wirelessly communicated between the UE **2** and eNB **11**. The eNB **11** can also communicate control plane data with the gNB **12**. Control plane data can propagate along the paths of the dashed lines in FIG. 4. The solid lines in FIG. 4 are for data plane paths.

[0122] In the example dual connectivity topology of FIG. 4, any suitable combinations of standardized bands and radio access technologies (e.g., FDD, TDD, SUL, SDL) can be wirelessly transmitted and received. This can present technical challenges related to having multiple separate radios and bands functioning in the UE **2**. With a TDD LTE anchor point, network operation may be synchronous, in which case the operating modes can be constrained to Tx1/Tx2 and Rx1/Rx2, or asynchronous which can involve Tx1/Tx2, Tx1/Rx2, Rx1/Tx2, Rx1/Rx2. When the LTE anchor is a frequency division duplex (FDD) carrier, the TDD/FDD inter-band operation can involve simultaneous Tx1/Rx1/Tx2 and Tx1/Rx1/Rx2.

[0123] As discussed above, EN-DC can involve both 4G and 5G carriers being simultaneously transmitted from a UE. Transmitting both 4G and 5G carriers in a UE, such as a phone, typically involves two power amplifiers (PAs) being active at the same time. Traditionally, having two power amplifiers active simultaneously would involve the placement of one or more additional power amplifiers specifically suited for EN-DC operation. Additional board space and expense is incurred when designing to support such EN-DC/NSA operation.

Examples of Features Supported by RFFE Systems Including Power Amplifiers

[0124] A radio frequency (RF) communication device can include multiple antennas for supporting wireless communications. Additionally, the RF communication device can include a radio frequency front-end (RFFE) system for processing signals received from and transmitted by the antennas. The RFFE system can provide a number of functions, including, but not limited to, signal filtering, signal partitioning and combining, controlling component connectivity to the antennas, and/or signal amplification.

[0125] RFFE systems can be used to handle RF signals of a wide variety of types, including, but not limited to, wireless local area network (WLAN) signals, Bluetooth® signals, and/or cellular signals. RFFE systems are also referred to herein as front-end systems.

[0126] RFFE systems can be used to process signals of a wide range of frequencies. For example, certain RFFE systems can operate using one or more low bands (for example, RF signal bands having a frequency content of 1 GHz or less, also referred to herein as LB), one or more mid bands (for example, RF signal bands having a frequency content between 1 GHz and 2.3 GHz, also referred to herein as MB), one or more high bands (for example, RF signal bands having a frequency content between 2.3 GHz and 3 GHz, also referred to herein as HB), and one or more ultrahigh bands (for example, RF signal bands having a frequency content between 3 GHz and 7.125 GHz, also referred to herein as UHB). In certain implementations, modules operate over mid band and high band frequencies (MHB).

[0127] Moreover, RFFE systems can also process millimeter wave frequency signals, such as those in FR2 of 5G.

[0128] RFFE systems can be used in a wide variety of RF communication devices, including, but not limited to, smartphones, base stations, laptops, handsets, wearable electronics, and/or tablets.

[0129] An RFFE system can be implemented to support a variety of features that enhance bandwidth and/or other performance characteristics of the RF communication device in which the RFFE system is incorporated.

[0130] For example, to support wider bandwidth, an increasing number of uplink carrier aggregation scenarios have been developed to support wider bandwidth. Additionally, the bandwidths for uplink and downlink cannot be arbitrarily sent since there is a minimum uplink bandwidth for maintaining a reliable link supported by the transport layer's ACK/NACK traffic. Thus, in 4G/5G, wideband uplink carrier aggregation should be supported to achieve higher bandwidth for downlink carrier aggregation.

[0131] Thus, an RFFE system can be implemented to support both uplink and downlink carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels, for instance up to five carriers. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

[0132] Transition from 4G to 5G is through non-standalone (NSA) operation, rather than directly to full standalone (SA) operation. Current networks operate in 4G and 5G concurrently by communicating with an eNodeB and a gNodeB simultaneously in an EN-DC mode of operation. Thus, 4G and 5G transmitters operate concurrently in such a phone.

[0133] To provide such feature support, an RFFE system can be implemented to support EN-DC.

[0134] Support for EN-DC can cover a wide range of frequency bands, including using a 4G band in the LB, MHB, HB, or UHB frequency ranges in combination with a 5G band in the LB, MHB, HB, or UHB frequency ranges. Thus, various combinations of EN-DC including, but not limited to, LB-LB EN-DC, MHB-MHB EN-DC, LB-MHB EN-DC, LB-UHB EN-DC, MHB-UHB EN-DC, and/or UHB-UHB EN-DC, are possible.

[0135] Moreover, in certain dual uplink transmission scenarios, it can be desirable to provide

flexibility between swapping which antenna transmits a first RF transmit signal (for instance, one of a 4G signal or a 5G signal) on a first side of a phone board assembly and which antenna transmits a second RF transmit signal (for instance, the other of the 4G signal or the 5G signal) on a side of the phone board assembly. To provide such flexibility, an RFFE system can support a transmit swap function to selectively switch which antenna a particular RF transmit signal is transmitted from.

[0136] Another technique for increasing uplink capacity is uplink multiple-input multiple-output (MIMO) communications, in which multiple (for instance, two) power amplifiers transmit two different signals simultaneously on the same frequency using different antennas. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. MIMO order refers to a number of separate data streams sent or received.

Overview of Example Embodiments for Biasing of Differential Power Amplifiers

[0137] As wireless communication systems evolve more and more frequency bands are integrated into a single RF front-end module with a shared power amplifier used to provide amplification to RF signals associated with multiple frequency bands. For example, in certain applications only one power amplifier is employed.

[0138] Furthermore, constraints on power amplifier linearity and efficiency continue to tighten with evolving radio communications specifications. Thus, it is desirable for a power amplifier to have linear gain versus output power and linear phase versus output power characteristics. For example, it is desirable for power amplifiers to have little to no amplitude distortion (AM/AM) and phase distortion (AM/PM).

[0139] Differential power amplifier architectures like push-pull amplifiers, Doherty amplifiers, or a combination of both (Doherty push-pull) can be used for power amplification in multi-band and/or high-performance applications.

[0140] Achieving a linear gain and phase characteristic while maintaining output power and efficiency is a key challenge in power amplifier design.

[0141] A power amplifier's bias circuit (also referred to herein as a bias network) plays an important role in determining the power amplifier's gain and phase distortion characteristics. For example, high linearity can be achieved by keeping the bias voltage constant with changes in input power.

[0142] Apparatus and methods for biasing of differential power amplifiers are disclosed herein. In certain embodiments, a power amplifier includes a first gain transistor for amplifying a first signal component of an RF signal and a second gain transistor for amplifying a second signal component of the RF signal. The power amplifier further includes a shared bias circuit for generating a bias signal for biasing an input of the first gain transistor and an input of the second gain transistor. The RF signal is coupled into the shared bias circuit to enhance biasing performance.

[0143] For example, by injecting the RF signal into the shared bias network, the bias signals maintain the input voltages to the gain transistors more constant with increasing RF signal power.

[0144] Moreover, by using a shared bias circuit, the area of biasing circuitry is more compact and die area is saved. The area reduction in the bias network can help other blocks grow in area to improve power amplifier performance, such as power amplifier efficiency. Furthermore, in certain implementations the shared bias circuit uses one or more virtual grounds to help in shrinking area while maintaining amplitude and phase distortion benefits (for instance, achieving performance like that of using separate bias networks for each side of the differential power amplifier).

[0145] The teachings herein are applicable to a wide range of differential power amplifier architectures including, but not limited to, Doherty power amplifier systems, push-pull power amplifier systems, and Doherty push-pull power amplifier systems.

[0146] For example, a Doherty power amplifier includes a main or carrier amplifier and an auxiliary or peaking amplifier that operate in combination with one another to amplify an RF

signal. The Doherty power amplifier combines a carrier signal from the carrier amplifier and a peaking signal from the peaking amplifier to generate an amplified RF output signal. In certain implementations, the carrier amplifier is enabled over a wide range of power levels by a class-AB bias circuit while the peaking amplifier is selectively enabled by a class-C bias circuit at high power levels. The class-AB bias circuit and/or the class-C bias circuit of a Doherty power amplifier can be implemented in accordance with one or more embodiments of the present disclosure.

[0147] FIG. 5A is a schematic diagram of a power amplifier **120** according to one embodiment. The power amplifier **120** includes an input balun **101**, an output balun **102**, a first amplification (g.sub.m) device **103**, a second amplification device **104**, a bias circuit **105**, a first input impedance **107**, a second input impedance **108**, an output termination impedance **109**, a coupling capacitor **113**, a first biasing impedance **115**, and a second biasing impedance **116**. The bias circuit **105** includes a follower device **111** that receives a reference signal REF.

[0148] In the illustrated embodiment, the input balun **101** includes a primary winding or side connected between an input terminal that receives an RF input signal RFIN and a ground voltage, and a secondary winding or side connected between a first end of the first input impedance **107** and a first end of the second input impedance **108**. The primary winding of the input balun **101** receives the RF input signal as a single-ended signal, and the secondary winding outputs a differential RF input signal including a first RF signal component and a second RF signal component. In certain implementations, the first RF signal component corresponds to a non-inverted signal component and the second RF signal component corresponds to an inverted signal component having a phase difference of about 180° relative to the non-inverted signal component. Circuitry of a power amplifier that amplifies the first RF signal component is referred to herein as an A-side (or A), while circuitry of the power amplifier that amplifies the second RF signal component is referred to herein as a B-side (or B).

[0149] Although the power amplifier **120** of FIG. 5A includes the input balun **101** for converting the single-ended RF input signal to a differential RF input signal, other circuitry (for instance, an input splitting network or other suitable circuitry) can be used to separate a single-ended RF input signal into multiple signal components.

[0150] With continuing reference to FIG. 5A, the bias circuit **105** generates a bias signal at an output that also serves as a virtual ground to the differential configuration of the first amplification device **103** and the second amplification device **104**. The first biasing impedance **115** includes a first end connected to the output of the bias circuit **105** and a second end connected to an input of the first amplification device **103**. The input of the first amplification device **103** is also connected to a second end of the first input impedance **107**. The second biasing impedance **116** includes a first end connected to the output of the bias circuit **105** and a second end connected to an input of the second amplification device **104**. The input of the second amplification device **104** is also connected to a second end of the second input impedance **108**.

[0151] In the illustrated embodiment, the bias circuit **105** includes a follower device **111** that receives a reference signal REF and that outputs the bias signal. The follower device **111** can include, for example, an emitter follower bipolar transistor or a source follower field-effect transistor (FET). Although the bias circuit **105** includes the follower device **111** in this embodiment, the teachings herein are applicable to other configurations of biasing circuitry.

[0152] The first amplification device **103** includes an output connected to first end of a primary winding of the output balun **102**, and the second amplification device **104** includes an output connected to a second end of the primary winding of the output balun **102**. The first amplification device **103** and the second amplification device **104** are also referenced to the ground voltage. The first amplification device **103** and the second amplification device **104** can be, for example, common emitter bipolar transistors or common source field-effect transistors.

[0153] With continuing reference to FIG. 5A, the output balun **102** further includes a secondary winding connected between an output terminal that provides an RF output signal RF.sub.OUT and

a first end of the output termination impedance **109**. The second end of the output termination impedance **109** is connected to the ground voltage, in this example.

[0154] As shown in FIG. 5A, the coupling capacitor **113** is connected between the input terminal and the output of the bias circuit **105**. The coupling capacitor **113** operates to couple the RF input signal RFIN into the output of the shared bias circuit **105** to enhance biasing performance. For example, by injecting the RF input signal RFIN into the shared bias circuit **105**, the input voltages to the first amplification device **103** and the second amplification device **104** are maintained more constant with increasing RF signal power. Thus, compensation to inhibit input voltage droop at high RF signal power is provided.

[0155] Moreover, injecting the RF input signal RFIN into the shared bias circuit **105** provides the power amplifier **120** with superior amplitude distortion and phase distortion characteristics relative to an implementation in which the RF input signal RFIN is not coupled into the shared bias circuit **105**.

[0156] FIG. 5B is a schematic diagram of a power amplifier **130** according to another embodiment. The power amplifier **130** includes an input balun **101**, an output balun **102**, a first amplification device **103**, a second amplification device **104**, a bias circuit **105**, a first input impedance **107**, a second input impedance **108**, an output termination impedance **109**, a coupling capacitor **113**, and a biasing inductor **114**.

[0157] The power amplifier **130** of FIG. 5B is similar to the power amplifier **120** of FIG. 5A, except that the power amplifier **130** of FIG. 5B omits the first biasing impedance **115** and the second biasing impedance **116** of FIG. 5A in favor of including the biasing inductor **114**.

[0158] As shown in FIG. 5B, the biasing inductor **114** is connected between the output of the shared bias circuit **104** and a center tap of the secondary winding of the input balun **101**. Thus, bias signals (for instance, bias currents) for both the first amplification device **103** and the second amplification device **104** are provided through the biasing inductor **114** to the inputs (for instance, bases or gates) of the amplification devices **103/104**.

[0159] FIG. 5C is a schematic diagram of a power amplifier **140** according to another embodiment. The power amplifier **140** includes an input balun **101**, an output balun **102**, a first amplification device **103**, a second amplification device **104**, a bias circuit **105**, a first input impedance **107**, a second input impedance **108**, an output termination impedance **109**, and a coupled biasing inductor **131**.

[0160] The power amplifier **140** of FIG. 5C is similar to the power amplifier **130** of FIG. 5B, except that the power amplifier **140** of FIG. 5C omits the coupling capacitor **113** and the biasing inductor **114** of FIG. 5B in favor of including the coupled biasing inductor **131**.

[0161] In the illustrated embodiment, the coupled biasing inductor **131** is coupled (for instance, magnetically or electromagnetically) to the input balun **101** (for instance, to the primary winding of the input balun **101**). The coupled biasing inductor **131** operates to couple the RF input signal RFIN into the output of the bias circuit **105**. The coupled biasing inductor **131** is electrically connected between the center tap of the secondary winding of the input balun **101** and the output of the bias circuit **105**, and serves to conduct bias signals (for instance, bias currents) provided from the bias circuit **105** to the inputs of the amplification devices **103/104**.

[0162] FIG. 6 is a schematic diagram of a power amplifier **150** according to another embodiment. The power amplifier **150** includes an input balun **101**, an output balun **102**, a first amplification device **103**, a second amplification device **104**, a bias circuit **145**, a first input impedance **107**, a second input impedance **108**, an output termination impedance **109**, a first coupling impedance **147**, and a second coupling impedance **148**. The bias circuit **145** includes a first follower device **111** and a second follower device **112** that receive a shared or common reference signal REF.

[0163] In the illustrated embodiment, the bias circuit **145** includes a first output that provides a first bias signal to an input of the first amplification device **103** through the first coupling impedance **147** and a second output that provides a second bias signal to an input of the second amplification

device **104** through the second coupling impedance **148**.

[0164] With continuing reference to FIG. **6**, the first amplification device **103** amplifies a first RF signal component of the RF input signal RFIN while the second amplification device **104** amplifies a second RF signal component of the RF input signal RFIN. Additionally, the first coupling impedance **147** couples a portion of the first RF signal component into the first output of the bias circuit **145**, while the second coupling impedance **148** couples a portion of the second RF signal component into the second output of the bias circuit **145**.

[0165] Such RF signal coupling enhances biasing performance. For example, by injecting the RF signal components into the outputs of the shared bias circuit **145**, the input voltages to the first amplification device **103** and the second amplification device **104** are maintained more constant with increasing RF signal power. Thus, compensation to inhibit input voltage droop at high RF signal power is provided.

[0166] FIG. **7** is a schematic diagram of a Doherty power amplifier system **260** according to one embodiment. The Doherty power amplifier system **260** includes an input terminal IN, a first DC blocking capacitor **201**, an input amplifier **202**, a second DC blocking capacitor **203**, an input bias inductor **204**, an input balun **205**, an input balun termination resistor **206**, a first input balun capacitor **207**, a second input balun capacitor **208**, a carrier amplifier **211**, a peaking amplifier **212**, a first inverter impedance **213**, a second inverter impedance **214**, an output balun **215**, an output termination capacitor **216**, and an output terminal OUT.

[0167] The Doherty power amplifier system **260** depicts an example of a power amplifier system that can be implemented with biasing in accordance with one or more embodiments of the present disclosure. Although the Doherty power amplifier system **260** depicts an example of a power amplifier system that can benefit from the biasing schemes disclosed herein, the teachings herein are applicable to a wide range of power amplifier systems.

[0168] In the illustrated embodiment, the input amplifier **202** receives an RF input signal from the input terminal IN through the first DC blocking capacitor **201**. An output of the input amplifier **202** receives a power amplifier supply voltage $V_{sub,CC}$ through the input bias inductor **204**. Additionally, the output of the input amplifier **202** provides an amplified RF input signal to the input balun **205** through the second DC blocking capacitor **203**.

[0169] The input balun **205** is connected with the depicted components to operate as an input signal separation circuit that separates the amplified RF input signal into a carrier input signal and a peaking input signal with a desired phase shift (for example, about ninety degrees in some implementations). In the illustrated embodiment, the input balun **205** includes a primary winding connected between the second DC blocking capacitor **203** and an input to the carrier amplifier **211**, and a secondary winding connected between an input to the peaking amplifier **212** and a first end of the input balun termination resistor **206**. The input balun termination resistor **206** further includes a second end electrically connected to ground. Additionally, the first input balun capacitor **207** and the second input balun capacitor **208** are each connected between the primary winding and the secondary winding at opposite ends of the input balun **205**.

[0170] In the illustrated embodiment, the carrier amplifier **211** includes a first inductor **221**, an input DC blocking capacitor **222**, a carrier driver amplifier **223**, a second inductor **224**, a balun **225**, a supply capacitor **226**, a first input capacitor **227**, a second input capacitor **228**, a class-AB bias circuit **229**, a first bipolar transistor **231**, and a second bipolar transistor **232**. The first inductor **221** is coupled between the input of the carrier amplifier **211** and ground. The carrier driver amplifier **223** receives the carrier input signal through the input DC blocking capacitor **222** and provides a single-ended carrier output signal to a first end of the primary winding of the balun **225** through the second inductor **224**. The primary winding of the balun **225** further includes a second end coupled to the power amplifier supply voltage $V_{sub,CC}$ and to a first end of the supply capacitor **226**. A second end of the supply capacitor **226** is electrically connected to ground. The secondary winding of the balun **225** includes a first end connected to the base of the first bipolar

transistor **231** through the first input capacitor **227**, and a second end connected to the base of the second bipolar transistor **232** through the second input capacitor **228**. The emitter of the first bipolar transistor **231** is connected to ground while a collector of the first bipolar transistor **231** is connected to a first end of the first inverter impedance **213**. The emitter of the second bipolar transistor **232** is connected to ground while a collector of the second bipolar transistor **232** is connected to a first end of the second inverter impedance **214**.

[0171] With continuing reference to FIG. 7, the peaking amplifier **212** includes a first inductor **241**, an input DC blocking capacitor **242**, a peaking driver amplifier **243**, a second inductor **244**, a balun **245**, a supply capacitor **246**, a first input capacitor **247**, a second input capacitor **248**, a class-C bias circuit **249**, a first bipolar transistor **251**, and a second bipolar transistor **252**. The first inductor **241** is coupled between the input of the peaking amplifier **212** and ground. The peaking driver amplifier **243** receives the peaking input signal through the input DC blocking capacitor **242** and provides a single-ended peaking output signal to a first end of the primary winding of the balun **245** through the second inductor **244**. The primary winding of the balun **245** further includes a second end coupled to the power amplifier supply voltage $V_{sub,CC}$ and to a first end of the supply capacitor **246**. A second end of the supply capacitor **246** is electrically connected to ground. The secondary winding of the balun **245** includes a first end connected to the base of the first bipolar transistor **251** through the first input capacitor **247**, and a second end connected to the base of the second bipolar transistor **252** through the second input capacitor **248**. The emitter of the first bipolar transistor **251** is connected to ground while a collector of the first bipolar transistor **251** is connected to a second end of the first inverter impedance **213**. The emitter of the second bipolar transistor **252** is connected to ground while a collector of the second bipolar transistor **252** is connected to a second end of the second inverter impedance **214**.

[0172] The output balun **215** includes a primary winding connected between the second end of the first inverter impedance **213** and the second end of the second inverter impedance **214**. The output balun **215** further includes a secondary winding connected between the output terminal OUT and a first end of the output termination capacitor **216**. The output termination capacitor **216** further includes a second end connected to ground.

[0173] The peaking amplifier **212** amplifies the peaking input signal and the carrier amplifier **211** amplifies the carrier input signal. The amplified RF peaking signal and the amplified RF carrier signal are combined using the impedance inverter to generate an RF output signal that is provided to the output terminal OUT.

[0174] As shown in FIG. 7, the peaking amplifier **212** includes the class-C bias circuit **249**, which biases the bases of the peaking amplifier's bipolar transistors **251/252**. Additionally, the carrier amplifier **212** includes the class-AB bias circuit **229**, which biases the bases of the carrier amplifier's bipolar transistors **231/232**. The class-C bias circuit **249** and/or the class-AB bias circuit **229** can be implemented in accordance with one or more embodiments of the present disclosure.

[0175] FIG. 8A is a schematic diagram of a power amplifier **330** according to another embodiment. The power amplifier **330** includes an input balun **301**, an input center tap capacitor **302**, a first input capacitor **303** (of capacitance CB_2), a second input capacitor **304** (of capacitance CB_2), a first biasing resistor **305** (of resistance RB_2), a second biasing resistor **306** (of resistance RB_2), a first bipolar transistor **311** (of area $A_{sub,E}$), a second bipolar transistor **312** (of area $A_{sub,E}$), an output balun **313**, an output center tap capacitor **314**, an output termination capacitor **315**, a bias circuit **316**, and a coupling capacitor **317**.

[0176] The power amplifier **330** of FIG. 8A is similar to the power amplifier **120** of FIG. 5A, except that the power amplifier of FIG. 8A includes specific implementations of components and circuits as shown in the figure.

[0177] In the illustrated embodiment, the amplification devices are implemented as bipolar transistors **311/312** each connected in a common emitter configuration. The bipolar transistors **311/312** have a nominal collector current I_{CQ} and a nominal base current $I_B = I_{CQ}/\beta$, wherein β is the

nominal bipolar transistor gain. In certain implementations, the bipolar transistors **311/312** are implemented as heterojunction bipolar transistors (HBTs).

[0178] The bias circuit **316** includes a reference current source **321** (generating a reference current IREF), a diode **322** (with area $A_{sub.E_EF/2M}$), a base-to-collector capacitor **323**, a base capacitor **324** (of capacitance $CB_{sub.EF}$), a collector capacitor **325** (of capacitance $CC_{sub.EF}$), a resistor **326** (of resistance $RB2 \times M/2$), an emitter follower bipolar transistor **327** (of area $A_{sub.E_EF}$), and a biasing bipolar transistor **328** (of area $A_{sub.E/2M}$).

[0179] As shown in FIG. **8A**, the base of the emitter follower bipolar transistor **327** receives a reference signal $I_B = I_C / \beta_{sup.2}$ at a base and outputs a bias signal from the emitter. The collector of the emitter follower bipolar transistor **327** is connected to a battery voltage $V_{sub.BATT}$. The bias signal is provided through the first biasing resistor **305** to the first bipolar transistor **311** and through the second biasing resistor **306** to the second bipolar transistor **312**.

[0180] As shown in FIG. **8A**, the RF input signal is coupled using the coupling capacitor **317** from the input terminal to the emitter of the emitter follower bipolar transistor **327**. The injection from the input terminal to the bias improves both amplitude distortion (AM/AM) and phase distortion (AM/PM) by keeping the base voltages of the first and second bipolar transistors **311/312** relatively constant with increasing RF signal power.

[0181] FIG. **8B** is a schematic diagram of a power amplifier **360** according to another embodiment. The power amplifier **360** includes an input balun **301**, an input center tap capacitor **302**, a first input capacitor **331** (of capacitance $CB2$), a second input capacitor **332** (of capacitance $CB2$), a first input resistor **333** (of resistance $RB2$), a second input resistor **334** (of resistance $RB2$), a first bipolar transistor **311**, a second bipolar transistor **312**, an output balun **313**, an output center tap capacitor **314**, an output termination capacitor **315**, a bias circuit **316**, a biasing inductor **357**, and a coupling capacitor **358**.

[0182] In comparison to the power amplifier **330** of FIG. **8A**, the power amplifier **360** of FIG. **8B** provides biasing from the output of the bias circuit **316** to the biasing inductor **357**, which is coupled to a center tap of the secondary winding of the input balun **301**. As shown in FIG. **8B**, the parallel combination of the first input capacitor **331** and the first input resistor **333** is provided between a first end of the secondary winding of the input balun **301** and a base of the first bipolar transistor **311**, while the parallel combination of the second input capacitor **332** and the second input resistor **334** is provided between a second end of the secondary winding of the input balun **301** and a base of the second bipolar transistor **312**. Furthermore, the coupling capacitor **358** is included between the input terminal and the output of the bias circuit **316**.

[0183] Thus, the depicted embodiment utilizes RF signal injection from the input terminal to the center tap feed of the input balun, while ballasting resistors **333/334** aid in keeping the base voltages of the first and second bipolar transistors **311/312** relatively constant with increasing RF signal power.

[0184] FIG. **8C** is a schematic diagram of a power amplifier **380** according to another embodiment. The power amplifier **380** includes an input balun **301**, an input center tap capacitor **302**, a first input capacitor **331** (of capacitance $CB2$), a second input capacitor **332** (of capacitance $CB2$), a first input resistor **333** (of resistance $RB2$), a second input resistor **334** (of resistance $RB2$), a first bipolar transistor **311**, a second bipolar transistor **312**, an output balun **313**, an output center tap capacitor **314**, an output termination capacitor **315**, a bias circuit **316**, and a coupled biasing inductor **375**.

[0185] The coupled biasing inductor **375** is coupled (for instance, magnetically or electromagnetically) to the input balun **301** (for instance, to the primary winding of the input balun **301**). The coupled biasing inductor **375** operates to couple the RF input signal $RF_{sub.IN}$ into the output of the bias circuit **316**. The coupled biasing inductor **375** is electrically connected between the center tap of the secondary winding of the input balun **301** and the output of the bias circuit **316**, and serves to conduct bias signals provided from the bias circuit **316** to the inputs of the

bipolar transistors **311/312**.

[0186] The power amplifier of FIG. **8C** is similar to the power amplifier **140** of FIG. **5C**, except that the power amplifier of FIG. **8C** includes specific implementations of components and circuits as shown in the figure.

[0187] In the illustrated embodiment, the amplification devices are implemented as bipolar transistors **311/312** each connected in a common emitter configuration. Additionally, the bias circuit **316** includes an emitter follower bipolar transistor **324**. The coupling inductor **375** provides coupling from the input terminal of the power amplifier **380** to the emitter of the emitter follower bipolar transistor **327**. The bias signal from the emitter follower bipolar transistor **327** flows through the coupling inductor **375**. A parallel combination of a base capacitor **331/332** and a ballasting resistor **333/334** are included at the bases of the bipolar transistors **311/312**, respectively.

[0188] FIG. **9A** is a schematic diagram of a power amplifier **410** according to another embodiment. The power amplifier **410** includes an input balun **301**, an input center tap capacitor **302**, a first input capacitor **303** (of capacitance $CB2$), a second input capacitor **304** (of capacitance $CB2$), a first biasing resistor **305** (of resistance $RB2$), a second biasing resistor **306** (of resistance $RB2$), a first bipolar transistor **311** (of area $A_{sub.E}$), a second bipolar transistor **312** (of area $A_{sub.E}$), an output balun **313**, an output center tap capacitor **314**, an output termination capacitor **315**, a bias circuit **406**, a first coupling capacitor **401**, and a second coupling capacitor **402**.

[0189] In comparison to the power amplifier **330** of FIG. **8A** that uses the coupling capacitor **317** for RF signal injection, the power amplifier **410** of FIG. **9A** includes the first coupling capacitor **401** and the second coupling capacitor **402**, which are connected between a first output and a second output of the bias circuit **406**, respectively, and the bases of the first bipolar transistor **311** and the second bipolar transistor **312**.

[0190] In the illustrated embodiment, the bias circuit **406** includes a reference current source **321** (generating a reference current I_{REF}), a diode **322** (with area $A_{sub.E_EF/2M}$), a base-to-collector capacitor **323**, a base capacitor **324**, a collector capacitor **325**, a resistor **326** (of resistance $RB2 \times M/2$), a first emitter follower bipolar transistor **327a**, a second emitter follower bipolar transistor **327b**, and a biasing bipolar transistor **328** (of area $A_{sub.E/2M}$).

[0191] As shown in FIG. **9A**, the bases of the emitter follower bipolar transistors **327a/327b** receive a common reference signal $I_B = I_C / \beta_{sup.2}$. The first emitter follower bipolar transistor **327a** outputs a first bias signal from the emitter, and the second emitter follower bipolar transistor **327b** outputs a second bias signal from the emitter.

[0192] Additionally, a parallel combination of the first boost capacitor **401** and the first biasing resistor **305** are included between the emitter of the first emitter follower bipolar transistor **327a** and the base of the first bipolar transistor **311**, and a parallel combination of a second boost capacitor **402** and a second biasing resistor **306** are included between the emitter of the second emitter follower bipolar transistor **327b** and the base of the second bipolar transistor **312**.

[0193] As shown in FIG. **9A**, the first boost capacitor **401** provides RF signal injection of the first RF signal component amplified by the first bipolar transistor **311** into the emitter of the first emitter follower bipolar transistor **327a**. Additionally, the second boost capacitor **402** provides RF signal injection of the second RF signal component amplified by the second bipolar transistor **312** into the emitter of the second emitter follower bipolar transistor **327b**.

[0194] FIG. **9B** is a schematic diagram of a power amplifier **420** according to another embodiment. The power amplifier **420** includes an input balun **301**, an input center tap capacitor **302**, a first input capacitor **303** (of capacitance $CB2$), a second input capacitor **304** (of capacitance $CB2$), a first biasing resistor **305** (of resistance $RB2$), a second biasing resistor **306** (of resistance $RB2$), a first bipolar transistor **311** (of area $A_{sub.E}$), a second bipolar transistor **312** (of area $A_{sub.E}$), an output balun **313**, an output center tap capacitor **314**, an output termination capacitor **315**, a bias circuit **416**, a first coupling capacitor **401**, and a second coupling capacitor **402**.

[0195] The power amplifier **420** of FIG. **9B** is similar to the power amplifier **410** of FIG. **9A**,

except that the power amplifier **420** of FIG. **9B** includes a different implementation of the bias circuit **416** that achieves smaller bias circuit area by scaling as well as removal of capacitors from the collectors and emitters of the emitter follower bipolar transistors.

[0196] For example, the bias circuit **416** includes a reference current source **321** (generating a reference current IREF), a diode **322'** (with area $A_{\text{sub.E_EF/M}}$), a base-to-collector capacitor **323**, a resistor **326** (of resistance $RB2 \times M/2$), a first emitter follower bipolar transistor **327a**, a second emitter follower bipolar transistor **327b**, and a biasing bipolar transistor **328** (of area $A_{\text{sub.E/M}}$). The base capacitor **324** and the collector capacitor **325** of the bias circuit **406** of FIG. **9A** are not included in the bias circuit **416** of FIG. **9B**.

[0197] With general references to FIGS. **10A-11B**, simulation results for various implementations of power amplifiers are shown. The graphs include plots in blue for a baseline case in which separate bias circuits are used for each side of the differential power amplifier. Using separate bias circuits results in a large die circuit area. The graphs include plots in red for results using a shared bias circuit for the indicated implementation scenario.

[0198] FIG. **10A** depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier **330** of FIG. **8A**.

[0199] FIG. **10B** depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier **360** of FIG. **8B**.

[0200] FIG. **10C** depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier **380** of FIG. **8C**.

[0201] FIG. **11A** depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier **410** of FIG. **9A**.

[0202] FIG. **11B** depicts examples of graphs of voltage and distortion characteristics versus output power for one implementation of the power amplifier **420** of FIG. **9B**.

[0203] FIG. **12** is a schematic diagram of one embodiment of a mobile device **800**. The mobile device **800** includes a baseband system **801**, a transceiver **802**, a front-end system **803**, antennas **804**, a power management system **805**, a memory **806**, a user interface **807**, and a battery **808**.

[0204] The mobile device **800** can be used communicate using a wide variety of communications technologies, including, but not limited to, 2G, 3G, 4G (including LTE, LTE-Advanced, and LTE-Advanced Pro), 5G NR, WLAN (for instance, Wi-Fi®), WPAN (for instance, Bluetooth® and ZigBee®), WMAN (for instance, WiMax®), and/or GPS technologies.

[0205] The transceiver **802** generates RF signals for transmission and processes incoming RF signals received from the antennas **804**. It will be understood that various functionalities associated with the transmission and receiving of RF signals can be achieved by one or more components that are collectively represented in FIG. **12** as the transceiver **802**. In one example, separate components (for instance, separate circuits or dies) can be provided for handling certain types of RF signals.

[0206] The front-end system **803** aids in conditioning signals transmitted to and/or received from the antennas **804**. In the illustrated embodiment, the front-end system **803** includes antenna tuning circuitry **810**, power amplifiers (PAS) **811**, low noise amplifiers (LNAs) **812**, filters **813**, switches **814**, and signal splitting/combining circuitry **815**. However, other implementations are possible.

[0207] For example, the front-end system **803** can provide a number of functionalities, including, but not limited to, amplifying signals for transmission, amplifying received signals, filtering signals, switching between different bands, switching between different power modes, switching between transmission and receiving modes, duplexing of signals, multiplexing of signals (for instance, diplexing or triplexing), or some combination thereof.

[0208] In certain implementations, the mobile device **800** supports carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier

aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

[0209] The antennas **804** can include antennas used for a wide variety of types of communications. For example, the antennas **804** can include antennas for transmitting and/or receiving signals associated with a wide variety of frequencies and communications standards.

[0210] In certain implementations, the antennas **804** support MIMO communications and/or switched diversity communications. For example, MIMO communications use multiple antennas for communicating multiple data streams over a single radio frequency channel. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. Switched diversity refers to communications in which a particular antenna is selected for operation at a particular time. For example, a switch can be used to select a particular antenna from a group of antennas based on a variety of factors, such as an observed bit error rate and/or a signal strength indicator.

[0211] The mobile device **800** can operate with beamforming in certain implementations. For example, the front-end system **803** can include amplifiers having controllable gain and phase shifters having controllable phase to provide beam formation and directivity for transmission and/or reception of signals using the antennas **804**. For example, in the context of signal transmission, the amplitude and phases of the transmit signals provided to the antennas **804** are controlled such that radiated signals from the antennas **804** combine using constructive and destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction. In the context of signal reception, the amplitude and phases are controlled such that more signal energy is received when the signal is arriving to the antennas **804** from a particular direction. In certain implementations, the antennas **804** include one or more arrays of antenna elements to enhance beamforming.

[0212] The baseband system **801** is coupled to the user interface **807** to facilitate processing of various user input and output (I/O), such as voice and data. The baseband system **801** provides the transceiver **802** with digital representations of transmit signals, which the transceiver **802** processes to generate RF signals for transmission. The baseband system **801** also processes digital representations of received signals provided by the transceiver **802**. As shown in FIG. 12, the baseband system **801** is coupled to the memory **806** to facilitate operation of the mobile device **800**.

[0213] The memory **806** can be used for a wide variety of purposes, such as storing data and/or instructions to facilitate the operation of the mobile device **800** and/or to provide storage of user information.

[0214] The power management system **805** provides a number of power management functions of the mobile device **800**. In certain implementations, the power management system **805** includes a PA supply control circuit that controls the supply voltages of the power amplifiers **811**. For example, the power management system **805** can be configured to change the supply voltage(s) provided to one or more of the power amplifiers **811** to improve efficiency, such as power added efficiency (PAE).

[0215] As shown in FIG. 12, the power management system **805** receives a battery voltage from the battery **808**. The battery **808** can be any suitable battery for use in the mobile device **800**, including, for example, a lithium-ion battery.

[0216] FIG. 13 is a schematic diagram of a power amplifier system **860** according to one embodiment. The illustrated power amplifier system **860** includes a baseband processor **841**, a transmitter/observation receiver **842**, a power amplifier (PA) **843**, a directional coupler **844**, front-end circuitry **845**, an antenna **846**, a PA bias control circuit **847**, and a PA supply control circuit **848**. The illustrated transmitter/observation receiver **842** includes an I/Q modulator **857**, a mixer **858**, and an analog-to-digital converter (ADC) **859**. In certain implementations, the transmitter/observation receiver **842** is incorporated into a transceiver.

[0217] The baseband processor **841** can be used to generate an in-phase (I) signal and a quadrature-phase (Q) signal, which can be used to represent a sinusoidal wave or signal of a desired amplitude, frequency, and phase. For example, the I signal can be used to represent an in-phase component of the sinusoidal wave and the Q signal can be used to represent a quadrature-phase component of the sinusoidal wave, which can be an equivalent representation of the sinusoidal wave. In certain implementations, the I and Q signals can be provided to the I/Q modulator **857** in a digital format. The baseband processor **841** can be any suitable processor configured to process a baseband signal. For instance, the baseband processor **841** can include a digital signal processor, a microprocessor, a programmable core, or any combination thereof. Moreover, in some implementations, two or more baseband processors **841** can be included in the power amplifier system **860**.

[0218] The I/Q modulator **857** can be configured to receive the I and Q signals from the baseband processor **841** and to process the I and Q signals to generate an RF signal. For example, the I/Q modulator **857** can include digital-to-analog converters (DACs) configured to convert the I and Q signals into an analog format, mixers for upconverting the I and Q signals to RF, and a signal combiner for combining the upconverted I and Q signals into an RF signal suitable for amplification by the power amplifier **843**. In certain implementations, the I/Q modulator **857** can include one or more filters configured to filter frequency content of signals processed therein.

[0219] The power amplifier **843** can receive the RF signal from the I/Q modulator **857**, and when enabled can provide an amplified RF signal to the antenna **846** via the front-end circuitry **845**.

[0220] The front-end circuitry **845** can be implemented in a wide variety of ways. In one example, the front-end circuitry **845** includes one or more switches, filters, duplexers, multiplexers, and/or other components. In another example, the front-end circuitry **845** is omitted in favor of the power amplifier **843** providing the amplified RF signal directly to the antenna **846**.

[0221] The directional coupler **844** senses an output signal of the power amplifier **823**.

Additionally, the sensed output signal from the directional coupler **844** is provided to the mixer **858**, which multiplies the sensed output signal by a reference signal of a controlled frequency. The mixer **858** operates to generate a downshifted signal by downshifting the sensed output signal's frequency content. The downshifted signal can be provided to the ADC **859**, which can convert the downshifted signal to a digital format suitable for processing by the baseband processor **841**.

Including a feedback path from the output of the power amplifier **843** to the baseband processor **841** can provide a number of advantages. For example, implementing the baseband processor **841** in this manner can aid in providing power control, compensating for transmitter impairments, and/or in performing digital pre-distortion (DPD). Although one example of a sensing path for a power amplifier is shown, other implementations are possible.

[0222] The PA supply control circuit **848** receives a power control signal from the baseband processor **841**, and controls supply voltages of the power amplifier **843**. In the illustrated configuration, the PA supply control circuit **848** generates a first supply voltage $V_{sub.CC1}$ for powering an input stage of the power amplifier **843** and a second supply voltage $V_{sub.CC2}$ for powering an output stage of the power amplifier **843**. The PA supply control circuit **848** can control the voltage level of the first supply voltage $V_{sub.CC1}$ and/or the second supply voltage $V_{sub.CC2}$ to enhance the power amplifier system's PAE.

[0223] The PA supply control circuit **848** can employ various power management techniques to change the voltage level of one or more of the supply voltages over time to improve the power amplifier's power added efficiency (PAE), thereby reducing power dissipation.

[0224] One technique for improving efficiency of a power amplifier is average power tracking (APT), in which a DC-to-DC converter is used to generate a supply voltage for a power amplifier based on the power amplifier's average output power. Another technique for improving efficiency of a power amplifier is envelope tracking (ET), in which a supply voltage of the power amplifier is controlled in relation to the envelope of the RF signal. Thus, when a voltage level of the envelope of the RF signal increases the voltage level of the power amplifier's supply voltage can be

increased. Likewise, when the voltage level of the envelope of the RF signal decreases the voltage level of the power amplifier's supply voltage can be decreased to reduce power consumption. [0225] In certain configurations, the PA supply control circuit **848** is a multi-mode supply control circuit that can operate in multiple supply control modes including an APT mode and an ET mode. For example, the power control signal from the baseband processor **841** can instruct the PA supply control circuit **848** to operate in a particular supply control mode.

[0226] As shown in FIG. **13**, the PA bias control circuit **847** receives a bias control signal from the baseband processor **841**, and generates bias control signals for the power amplifier **843**. In the illustrated configuration, the bias control circuit **847** generates bias control signals for both an input stage of the power amplifier **843** and an output stage of the power amplifier **843**. However, other implementations are possible.

Applications

[0227] Some of the embodiments described above have provided examples in connection with mobile devices. However, the principles and advantages of the embodiments can be used for any other systems or apparatus that have needs for power amplifier systems. Examples of such RF communication systems include, but are not limited to, mobile phones, tablets, base stations, network access points, customer-premises equipment (CPE), laptops, and wearable electronics.

CONCLUSION

[0228] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

[0229] Moreover, conditional language used herein, such as, among others, “may,” “could,” “might,” “can,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0230] The above detailed description of embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed at

different times.

[0231] The teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

[0232] While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A mobile device comprising: a transceiver configured to generate a radio frequency input signal; and a front-end system including a power amplifier that includes a first amplification device configured to amplify a first signal component of a radio frequency input signal, a second amplification device configured to amplify a second signal component of the radio frequency input signal, a shared bias circuit having a first output configured to generate a first bias signal and a second output configured to generate a second bias signal, a first coupling impedance connected between the first output of the shared bias circuit and an input of the first amplification device, and a second coupling impedance connected between the second output of the shared bias circuit and an input of the second amplification device, the first coupling impedance configured to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output, and the second coupling impedance configured to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.
2. The mobile device of claim 1 wherein the power amplifier further includes an input terminal configured to receive the radio frequency input signal.
3. The mobile device of claim 2 wherein the power amplifier further includes an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device.
4. The mobile device of claim 3 wherein the power amplifier further includes a first capacitor connected between a first end of the secondary winding of the input balun and the input of the first amplification device, and a second capacitor connected between a second end of the secondary winding of the input balun and the input of the second amplification device.
5. The mobile device of claim 1 wherein the shared bias circuit generates the first bias signal and the second bias signal based on a common reference signal.
6. The mobile device of claim 1 wherein the first coupling impedance includes a first resistor and a first capacitor connected in parallel between the first output of the shared bias circuit and the input of the first amplification device, and the second coupling impedance includes a second resistor and a second capacitor connected in parallel between the second output of the shared bias circuit and the input of the second amplification device.
7. The mobile device of claim 1 wherein the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared bias circuit.
8. The mobile device of claim 7 wherein a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.
9. The mobile device of claim 1 wherein the first amplification device includes a first bipolar

transistor having a base corresponding to the input of the first amplification device, and the second amplification device includes a second bipolar transistor having a base corresponding to the input of the second amplification device.

10. The mobile device of claim 1 wherein the front end system includes a Doherty power amplifier system including the power amplifier.

11. The mobile device of claim 10 wherein the power amplifier is one of a peaking amplifier or a carrier amplifier of the Doherty power amplifier system.

12. A power amplifier comprising: a first amplification device configured to amplify a first signal component of a radio frequency input signal; a second amplification device configured to amplify a second signal component of the radio frequency input signal; a shared bias circuit having a first output configured to generate a first bias signal and a second output configured to generate a second bias signal; a first coupling impedance connected between the first output of the shared bias circuit and an input of the first amplification device, the first coupling impedance configured to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output; and a second coupling impedance connected between the second output of the shared bias circuit and an input of the second amplification device, the second coupling impedance configured to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.

13. The power amplifier of claim 12 further comprising an input terminal configured to receive the radio frequency input signal.

14. The power amplifier of claim 13 further comprising an input balun having a primary winding connected to the input terminal, and a secondary winding connect between the input of the first amplification device and the input of the second amplification device.

15. The power amplifier of claim 12 wherein the first coupling impedance includes a first resistor and a first capacitor connected in parallel between the first output of the shared bias circuit and the input of the first amplification device, and the second coupling impedance includes a second resistor and a second capacitor connected in parallel between the second output of the shared bias circuit and the input of the second amplification device.

16. The power amplifier of claim 12 wherein the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared bias circuit.

17. The power amplifier of claim 16 wherein a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.

18. A method of radio frequency signal amplification in a mobile device, the method comprising: amplifying a first signal component of a radio frequency input signal using a first amplification device of a power amplifier; amplifying a second signal component of the radio frequency input signal using a second amplification device of the power amplifier; generating a first bias signal at a first output of a shared bias circuit and a second bias signal at a second output of the shared bias circuit; using a first coupling impedance to provide the first bias signal to the input of the first amplification device and to couple the first signal component into the first output; and using a second coupling impedance to provide the second bias signal to the input of the second amplification device and to couple the second signal component into the second output.

19. The method of claim 18 wherein the shared bias circuit includes a first emitter follower bipolar transistor having an emitter connected to the first output of the shared bias circuit, and a second emitter follower bipolar transistor having an emitter connected to the second output of the shared bias circuit.

20. The method of claim 19 wherein a base of the first emitter follower bipolar transistor and a base of the second emitter follower bipolar transistor receive a common reference signal.

21.-50. (canceled)

